

Challenge 7 – Student/Roll No.: G VENKATA GANESH/192425381

<pre> 46 / \ 23 69 / \ / \ 11 34 58 81</pre>	A. Binary Tree (4 marks): Write the preorder, inorder, postorder and level-order traversals of the given tree. B. BST (3 marks): Insert the keys in this order: 46, 23, 69, 12, 35, 57, 80. Draw the final BST and write its postorder traversal. C. AVL Tree (3 marks): Insert 50, 20, 30 into an empty AVL tree. Name the rotation, draw the balanced tree and write its preorder traversal.
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A. Binary Tree – Traversals

Given tree:

```
      46
     /  \
    23   69
   / \  / \
  11 34 58 81
```

1. Preorder (Root → Left → Right):

46, 23, 11, 34, 69, 58, 81

2. Inorder (Left → Root → Right):

11, 23, 34, 46, 58, 69, 81

3. Postorder (Left → Right → Root):

11, 34, 23, 58, 81, 69, 46

4. Level-order:

46, 23, 69, 11, 34, 58, 81

B. BST

Insert keys in this order:

46, 23, 69, 12, 35, 57, 80

Final BST:

```
      46
     /  \
    23   69
   / \  / \
  12 35 57 80
```

BST Inorder Traversal:

12, 23, 35, 46, 57, 69, 80

C. AVL Tree

Insert:

50, 20, 30

Before balancing:

```
    50
   /
  20
   \
   30
```

This creates an **LR (Left-Right) imbalance** at 50.

Rotation:

1. Left rotation on 20
2. Right rotation on 50

Final balanced AVL tree:

```
    30
   /  \
  20   50
```

Preorder Traversal:

30, 20, 50